

The benefits of working with few items at a time | Agile Insider

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The Benefits of Working with Few Items at a Time What the effects of high WIP (work in process) can cause on the work system and how to manage it to avoid the negative effects. The objective of this text is to present a more accurate view of the management of the amount of activities that we have in the teams. Considering this, the following paragraphs explore how the teams are impacted and consequently the impact on the organizations when there is not good management of the amount of activities. The content approached here is pertinent to any area of work, especially where there is constant use of cognitive abilities and creative skills. Thus, accounting, administrative, marketing, technology among other areas, can apply the proposed concepts.

Main topics

- What WIP (Work in Process) is
- What the effects of high WIP can cause on the work system
- How to manage the WIP to avoid the negative effects
- How to deliver a work item faster?
- How to avoid the team overload?
- How to improve the quality of the team output?
- And even, how to deliver more?

These are some examples of very common challenges in teams that are in constant evolution, be this for opportunity or for necessity. There are innumerable strategies, tools, actions of different natures to respond to those challenges. Some measures could be to hire new professionals to the team, organize workshops, implant new tools, work on big projects etc. However, all of these items have high costs and long duration of time, depending on many sponsors, and can unfold good results in the long term or can unfold no results. I propose something simpler. Let's look at a no cost strategy to implement which depends almost exclusively on you and your team — I refer to work in progress management, or also known as WIP (Work in Process). When quantified, WIP is the amount of work items that started and not are finished yet, in other words, the amount of items that are in the workflow. In the case of a software development team, for example, the features that are being worked on at a certain moment. In the case of an accounting team, these items could be the accounting reports. We can also consider that the work items can be projects which are being worked on by several teams. In this context, the management of the amount of projects in progress in a department or even in the whole company, has a bigger importance. If done in a correct way, the company can considerably decrease the time-to-market, among others benefits approached later on.

Comparing with the retailer

Before exploring the ways of having better WIP management, let's make an analogy, imagine each activity in your work system is like a product in the inventory. A good inventory turnover is one of the most important factors for a retailer. Imagine the commercial area in a retail company made a considerable mistake in the strategy and bought a much bigger quantity than the ideal from suppliers. Over several days, a lot of truck drivers arrived at the distribution center of this retailer with all of these products. You can imagine that this company would start to have many problems such as:

- Big quantity of immobilized capital in stock — this extra expense that was used to buy products could have been used for other emergencies of the company (less capacity to react to changes);
- It becoming harder to handle the product in the distribution center, resulting in an increased number of damaged products (wastage);
- After some months, the customers would lose the interest in these products, they desire new models, so the part of stock which is stagnant would have to be sold below the cost price (obsolescence);
- With the full storage capacity, the retailer would delay to get reserve space for new launches awaited by the market (lower capacity of reacting to changes);
- The difficulty in handling the products would delay the shipment of part of the products to the customers (slowness);
- With so many trucks arriving, the distribution center entrance could become the bottleneck, in a short time a long line of trucks formed.

In addition, the average download time leaped, this would cause great stress on employees

and truck drivers (slowness and overload). The effects Despite not handling stocks, such workers in other areas as previously mentioned, can suffer even worse effects than this retailer, if working with too many items simultaneously. Again, let's take the example of a team that creates software and verify what these effects are: Focus switching— with multiple items in progress, it's common for team members to constantly switch the focus activity. This context switch consumes a lot of mental energy, especially in activities that demand high concentration, such as software development. The focus recovery is not immediate, it takes a long time and many of the problems (bugs) are generated unnoticed at these moments. Coordination cost— when a team is working on multiple projects at the same time, there is a huge amount of communication and information that needs to flow everywhere. This demands more meetings, more status reports, more presentations etc. In these situations, it's common to see managers with extremely busy schedules. This implies another problem, the manager's lack of time for issues related to the team and people's development. Capability of reacting to change— teams that are working with high WIP have a slower capability to change the direction. For example, if there are many projects running in parallel and a sudden change in the company's strategies has emerged (due to market factors or diverse factors, such as a pandemic) it will take a long time to finish the projects that are in progress and start new ones, those that represent the company's reaction to the change. Risk of wastage— a team that is working with high WIP and is faced with the need to discard these items to start another objective, will throw all this work (partially completed) in the trash. The greater the number of items in progress, the greater the risk of wastage. This is also correlated with how long these items remain in the workflow, we will address this time factor later. Just as the stuck product that in the stock has not yet fulfilled its mission (to be sold), partially completed work is an investment that has not yet paid off. In the case of software development teams, for instance, the new features being built will only deliver value when the clients can take advantage of them. So if these items are discarded before reaching the proposed value, the time dedicated by the individuals is wasted. Instead of having 10 items in progress and none completed, it is preferable to have 4 finished items that have delivered value and another 4 in progress. Quality— there are some cases that indicate the correlation between the high amount of work in progress and the bad quality of team output. Teams that start to work with less WIP demonstrate improvement in the quality output. My hypothesis is this negative effect on quality is due to switching activities mentioned above and the workload that high WIP generates. The effects mentioned above would be enough to decrease your team's WIP. However, the most relevant negative effect by high WIP, is in the period of time that an item takes to go through the workflow, this period is known as Cycle Time. The higher the WIP, the longer time it will take for an item to go through the flow. There are other factors besides high WIP that also impact the Cycle Time. We won't go into these details, because the factors and effects generated in the work system give a good reflection by itself. What do we need to start working on WIP? Well, I imagine you've already noticed that we need to work with a smaller amount of items at a time. But before that, it is necessary to create some conditions so that we can work on WIP: Visual management— different from the retailer where the stock is totally visible (number of boxes in the distribution center), the work done by teams based on cognitive abilities, are not palpable and visible. What is not seen is not managed and therefore it is not possible to make any kind of improvement. For that reason, whether on the company wall or in any activities management tool (in Luizalabs we use the SwiftKanban) it is necessary to bring up all the items the team is working on. Just by taking a look, the image below already demonstrates the WIP amount in the whole system and how these items are distributed in the work steps. Structured flow— it is also necessary to understand with the team what are the steps of the workflow. This sequence of steps already exists in any work process, just make them visible to everyone. To facilitate this exercise, I recommend the use of STATIK, a technique based on systemic view that facilitates the understanding of

the flow of producing valuable work. In this flow, it is necessary to make the stages that the items are not being worked on visible, they will be waiting for a considerable amount of time before someone takes action — these steps are called queues. Some examples of queues steps are shown in the below image: (Wait) To review, Waiting for Homologation and Waiting for Deploy. Working the WIP

Ok, now that we've prepared the ground with visual management and structured flow, we can discuss the mechanism to limit WIP, this is to restrict the maximum number of items at a time in the workflow, avoiding the effects explained above. Almost always, this involves decreasing the number of items in the flow that the team is used to dealing with. Normally, people have the first negative feeling when we propose working with only a few items at a time. Due to habit, we have the impression that we are more productive when we are doing "a thousand things at the same time". We always hear someone say things like: "I'm up to my ears with things to do". Really, it's something different from what we are used to, it's counter intuitive. The movement to limit the WIP should be very well explained and in accordance not only with the team, but also with the stakeholders. It's important to clarify the whys of this practice and its benefits; something this text attempts to do. There's not an exact recipe to limit the WIP. It depends on the various factors which compose the context of each team. However, there are some practices that we learn in the literature and combined with experience in dozens of teams in the last years. Considerations: Do not impose the WIP limit in an individual decision, together with all members of the team, agree on the number. The number of WIP should be calibrated over time, in other words, the team can decrease or increase this number according to the learnings. An abrupt change to a very low WIP can cause excessive stress in the process and generate resistance. Leave it clear to all stakeholders (product owners, key users, among others) that the team is working with limited WIP, how much is this WIP (ex.: 6 features at a time), the importance of respecting this and the benefits expected. This practice will only generate benefits if the team keeps the discipline of respecting the WIP over time. Step by step: Just start— establish a value of maximum WIP to the whole board. Example: 2 times the amount of people in the team. If there are 8 people in the team, start with a maximum WIP of 16. Or, the number of people in the team plus 1, if there are 8 people in the team, start with a maximum WIP of 9. Respect the WIP— put into practice a pull system, in other words, just pull a new demand when there is space. Monitor the work system— have bottleneck steps emerged? Are the blocked items clear? Have the metrics changed? Later we will talk about the CFD chart which assists this monitoring. Create an environment where people collaborate— how do we remove these bottlenecks? How do we reduce the blocks? What can we adjust in the work system? In addition to limiting the WIP, there are other actions which help take the pressure off having so many items in the work system, such as: There is good work to be done on the items before they come into the team focus, the manner that just well-cut items arrive (refinement) and that will really give value (triage). Work to reduce the volume of operational items and problems (bugs) to "leave" more space for items that will add value. Mentioning this, in a way, is obvious, however we do not always have a good idea of the large number of activities which consume space (time and energy) in the precious and scarce capacity of the team. Monitoring the WIP

On the line of monitoring the amount of WIP and the work system over time, I propose to utilize the CFD (cumulative flow diagram) chart. The CFD shows the cumulative amount of items in each of the stages over the weeks. Each "layer" of the chart represents one step. In the above example, every step of the flow was summarized in a macro step called WIP (red layer), so, we can follow if the red layer is more slim or more wide. Keep the layer slim, in other words, keep the WIP in a small amount and constant for weeks, depending on the items amounts that entered in the process and the number of finished items. For example, from week 4 to 5 in the above chart, one item entered and none exited (done), therefore the WIP increased. Previously we mentioned the considerable influence of the WIP in the Cycle Time (the higher the WIP, the longer the Cycle Time). The CFD chart makes it very clear, between week 2 and 4, when the

WIP was very contained, we noticed the average Cycle Time (represented by the horizontal arrow) much smaller than the average Cycle Time of weeks 4 and 8, given that in these weeks the WIP increased. In addition to monitoring the WIP (red layer) and the average Cycle Time, the CFD is very useful to track the amount of items in team backlog, the amount of items done and, thereby, can also be used to forecast how many items can be finished in the future based on the history of previously done items. Collaboration I have finished my activity, which one shall I start now? This is a very common phrase in the day-to-day of teams. Once we start respecting the maximum WIP, this question should change to something like: I have finished my activity, which other one can I help someone to finish before starting a new one? If the team already has the maximum number of activities in progress agreed, this is the moment to help other people to finish some activity. When I say “finish” in a generic way, I mean any activity. In software development teams, for example, at a moment like this, the developer “can make himself available” to help with the publishing in a production environment. As previously described, while not finished, an activity does not deliver value. This is one of the triggers for the “stop starting, start finishing” transition. Working with limited WIP is an intense dose of stimulation for collaboration among the people in the team. The whole that the team delivers overlaps the individual deliveries. Final The practice of limiting work in process goes far from being a stand-alone thing in the workflow management, it's interlaced with several other concepts and practices that have an immense potential to the teams, such as: queues, bottlenecks, flow efficiency, work item size, variability, among others. However, this is a powerful mechanism to bring immediate benefits to the team, also to avoid overload and its consequences on the people and work system. It's necessary to have discipline to say “no” to new demands when we already have the maximum WIP. This “no” is a negative that elevates the importance to correct prioritization and refinement of items that will be available to enter the workflow. Furthermore, on the line to bring positive discussions about the problems in the organization, David Anderson defends that the limited WIP generates “positive tension”: The discussion and collaboration invoked for the positive tension of limited WIP are healthy. It's a mechanism that enables the emergence of a continuous improvement culture. Without a limit to WIP, the progress on the improvement process is slow. References Kanban: Successful Evolutionary Change for your Technology Business: Successful Evolutionary Change for your Technology — David J. Anderson The Principles of Product Development Flow: Second Generation Lean Product Development — Donald G. Reinertsen Zen and the Software Management Art (Software Zen) — Alisson Vale